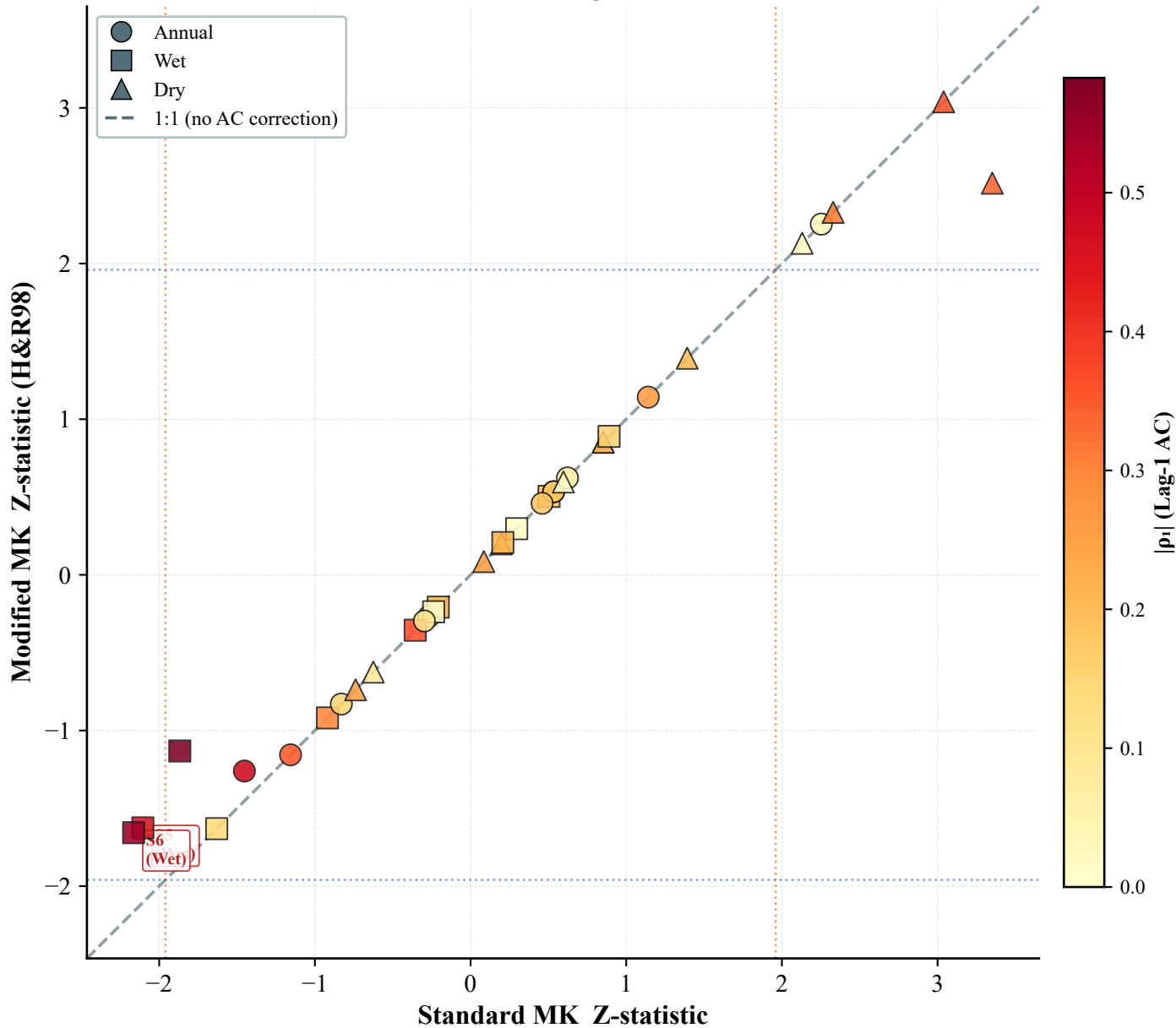
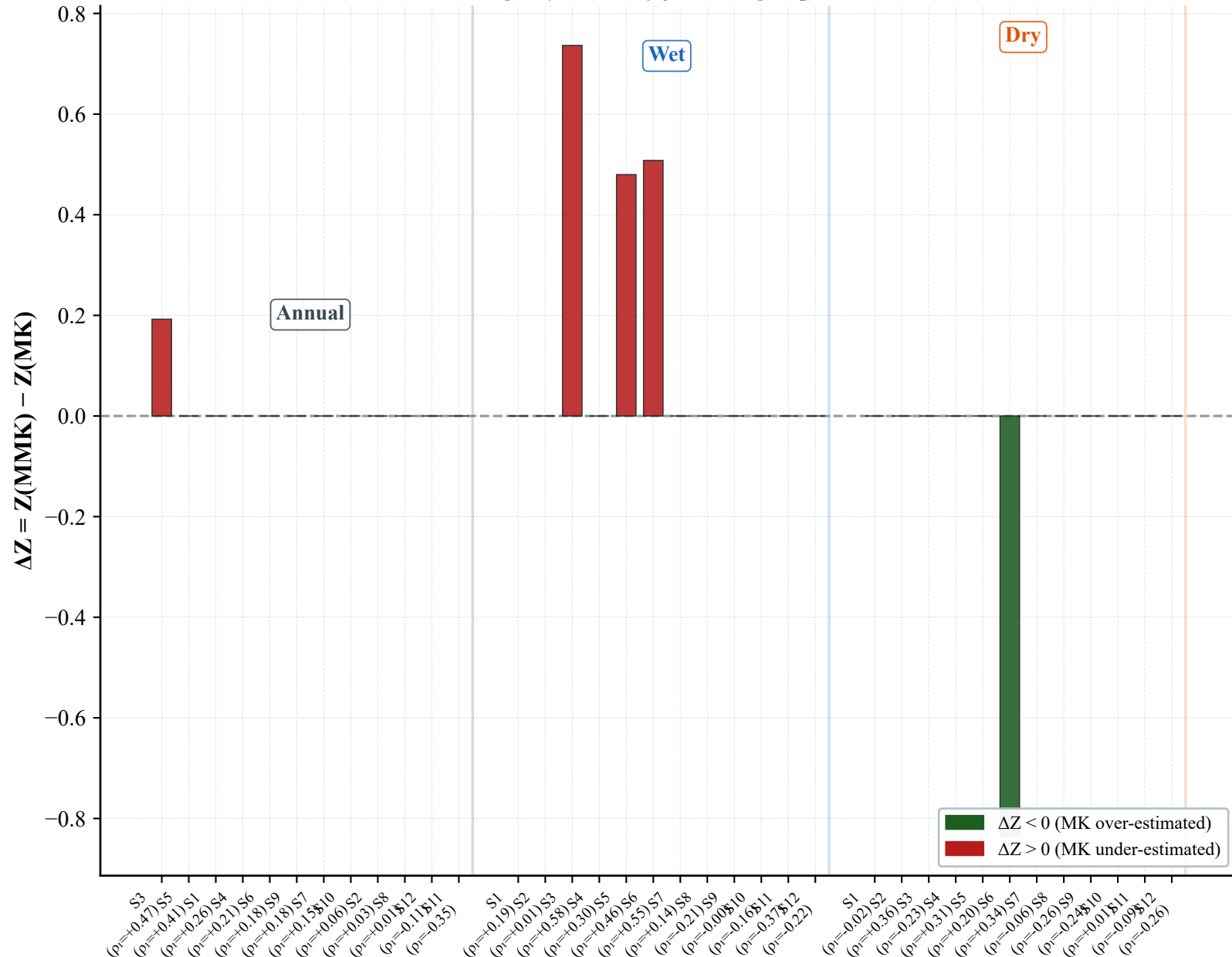
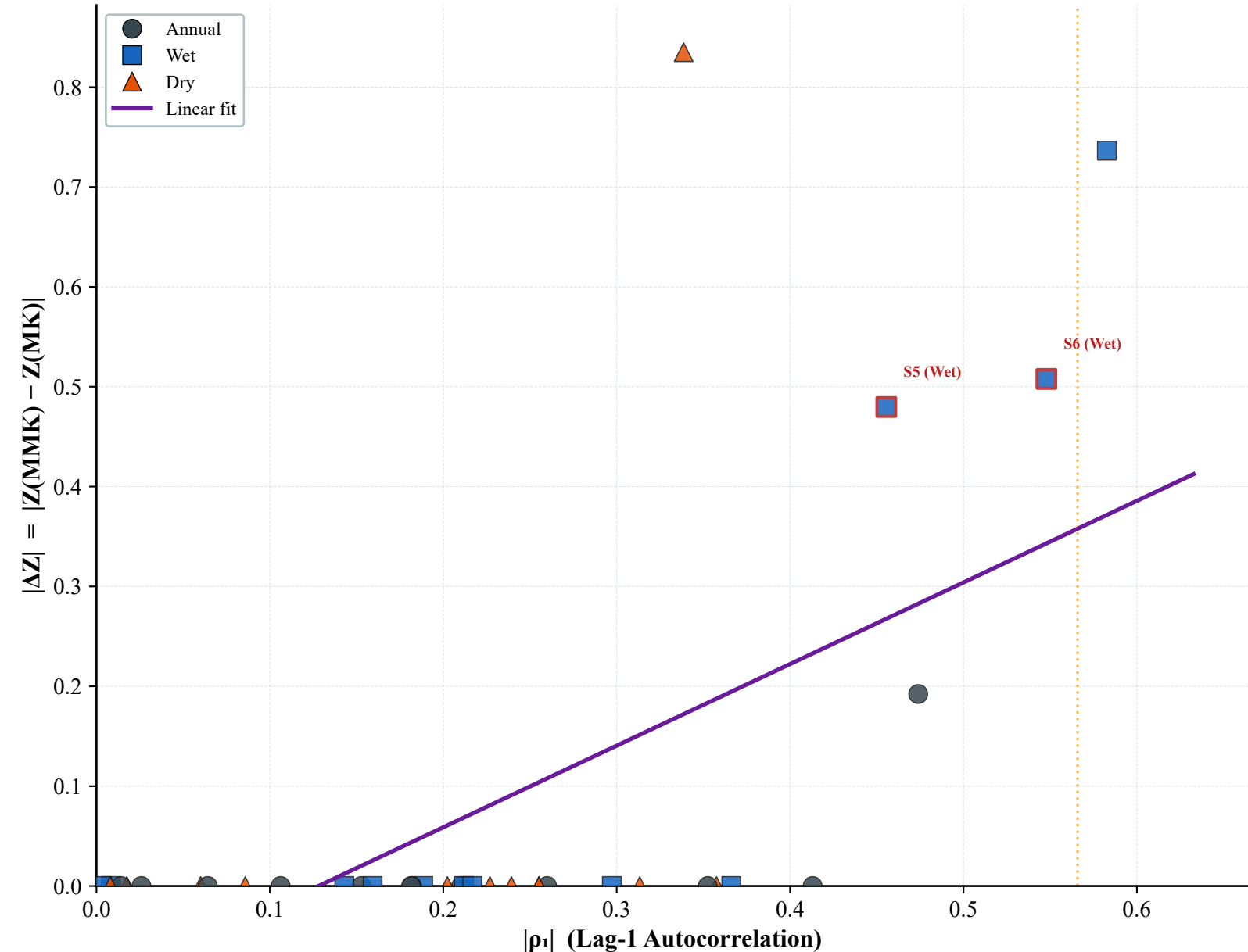


Figure 10. Methodological Implication: Autocorrelation Effect on MK vs MMK Trend Detection

(a) Standard MK vs Modified MK — Z-statistic Comparison

Study Period: 1981–2014 | Stations with significant ρ_1 (annual): 3/12 | Cases where AC changed trend conclusion: 2
Colour = $|\rho_1|$ | Points above 1:1 → AC inflated MK Z
Red labels = cases where AC correction changed trend conclusion(b) $\Delta Z = Z(\text{MMK}) - Z(\text{MK})$ per Station $\times$ ScaleReference: Hamed & Rao (1998) J. Hydrol. 204:182–196
Negative ΔZ → autocorrelation inflated Standard MK
= cases where trend conclusion changed | Sorted by ρ_1 within group

(c) Dose–Response: Autocorrelation Magnitude vs Z-Correction

Higher $|\rho_1|$ → larger deviation between MK and MMK Z
Red outline = trend conclusion changed after MMK correction

(d) Annual-Scale Summary: Autocorrelation Effect on MK vs MMK

Orange row = significant AC (ρ_1 significant) | Red row = trend conclusion changed after correction n_{eff} = effective sample size (Hamed & Rao 1998) | $\Delta Z = Z(\text{MMK}) - Z(\text{MK})$

Station	ρ_1	Sig.AC?	n	n_{eff}	Z (MK)	Z (MMK)	ΔZ	MK Trend	MMK Trend	Conclusion
S1	+0.260	No	34	34.0	1.142	1.142	+0.000	No trend	—	Consistent
S2	+0.026	No	34	34.0	2.253	2.253	+0.000	Increasing ↑	—	Consistent
S3	+0.474	Yes***	34	25.6	-1.453	-1.260	+0.192	No trend	—	AC corrected
S4	+0.210	No	34	34.0	0.534	0.534	+0.000	No trend	—	Consistent
S5	+0.413	Yes***	34	34.0	-0.237	-0.237	+0.000	No trend	—	AC corrected
S6	+0.182	No	34	34.0	0.534	0.534	+0.000	No trend	—	Consistent
S7	+0.153	No	34	34.0	-0.830	-0.830	+0.000	No trend	—	Consistent
S8	+0.014	No	34	34.0	-0.267	-0.267	+0.000	No trend	—	Consistent
S9	+0.181	No	34	34.0	0.460	0.460	+0.000	No trend	—	Consistent
S10	+0.064	No	34	34.0	0.623	0.623	+0.000	No trend	—	Consistent
S11	-0.352	Yes***	34	34.0	-1.156	-1.156	+0.000	No trend	—	AC corrected
S12	-0.106	No	34	34.0	-0.296	-0.296	+0.000	No trend	—	Consistent